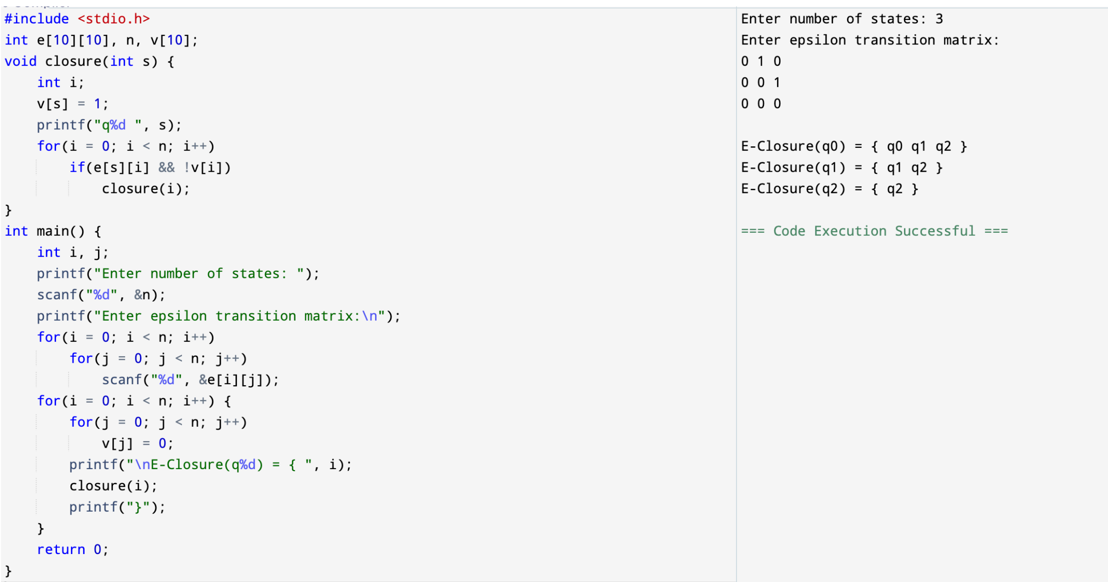
10. Write a C program to find ε -closure for all the states in a Non-Deterministic Finite Automata (NFA) with ε -moves.

Code:

```
#include <stdio.h>
int e[10][10], n, v[10];
void closure(int s) {
   int i;
   v[s] = 1;
   printf("q%d ", s);
   for(i = 0; i < n; i++)
      if(e[s][i] && !v[i])
         closure(i);
}
int main() {
   int i, j;
   printf("Enter number of states: ");
   scanf("%d", &n);
   printf("Enter epsilon transition matrix:\n");
   for(i = 0; i < n; i++)
      for(j = 0; j < n; j++)
         scanf("%d", &e[i][j]);
   for(i = 0; i < n; i++) {
      for(j = 0; j < n; j++)
         v[j] = 0;
      printf("\nE-Closure(q%d) = { ", i);
      closure(i);
      printf("}");
   }
   return 0;
}
```

Output:



```
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int e[10][10], n, v[10];
void closure(int s) {
    int i;
    v[s] = 1;
    printf("q%d ", s);
    for(i = 0; i < n; i++)
        if(e[s][i] && !v[i])
            closure(i);
}
int main() {
    int i, j;
    printf("Enter number of states: ");
    scanf("%d", &n);
    printf("Enter epsilon transition matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < n; j++)
            scanf("%d", &e[i][j]);
    for(i = 0; i < n; i++) {
        for(j = 0; j < n; j++)
            v[j] = 0;
        printf("\nE-Closure(q%d) = { ", i);
        closure(i);
        printf("}");
    }
    return 0;
}
```

```
Enter number of states: 3
Enter epsilon transition matrix:
0 1 0
0 0 1
0 0 0

E-Closure(q0) = { q0 q1 q2 }
E-Closure(q1) = { q1 q2 }
E-Closure(q2) = { q2 }

=== Code Execution Successful ===
```